

Final Project Decomposition: Restaurant Analysis

Research Area

Focus: Restaurant Analysis

Tables Used: restaurant, orders

Objective

To identify which restaurants are most popular, which generate the highest revenue, and explore why. Insights will be presented in a visual dashboard with actionable business recommendations based on trends in sales, ratings, cost, and city-based distribution.

Key Business Questions

1. What restaurants are generating the most revenue?
2. Which restaurants have the highest number of orders?
3. Are higher-rated restaurants associated with higher revenue?
4. How does restaurant performance vary across cities?
5. Do average meal costs affect popularity or revenue?
6. Which cuisines are most represented among high-performing restaurants?

Hypotheses

- Restaurants with more orders have higher revenue.
- Higher-rated restaurants generate more revenue.
- Cities with more restaurants drive higher total sales.
- There is a positive relationship between average cost and revenue.
- Certain cuisines dominate in both popularity and profitability.

Data Preprocessing & Cleaning Plan

1. Join Tables

- Join `orders` and `restaurant` tables using `restaurant_id` and `id`.

2. Clean & Transform Columns

- Convert `order_date` to proper date format.
- Filter for relevant currency (if needed).
- Create derived columns:
 - `month`, `year` from `order_date`
 - `total_sales` by summing `sales_amount`
 - `total_orders` by counting `sales_qty`

3. Simplify Categorical Columns

- Clean or simplify `cuisine` if needed (optional — time permitting)
- Use only `city` instead of full `address` to avoid excessive data cleaning.

Key Metrics & Parameters

- **Total Revenue per Restaurant**
- **Total Orders per Restaurant**
- **Average Sales per Order**
- **Average Rating per Restaurant**
- **Rating Count**
- **Average Cost of Meal per Restaurant**
- **Total Revenue per City**
- **Average Revenue by Rating Tier**
- **Top Performing Restaurants by Cuisine**

Visualizations

- **Bar Charts**
 - Top 10 Restaurants by Revenue
 - Top 10 Restaurants by Orders
 - Average Cost vs Revenue
 - Ratings vs Revenue
- **Pie Chart or Donut Chart**
 - Share of Total Revenue by City
 - Cuisine distribution among top restaurants
- **Heatmap or Matrix**
 - Correlation of Rating, Cost, and Revenue

- **Line or Area Chart (if time permits)**
 - Monthly Revenue Trends (aggregate over time)
- **Scatter Plot**
 - Rating vs Revenue (with cost as color or size)

Story Flow

The dashboard will tell the following story:

1. **Overview:** High-level revenue and order performance by restaurant.
2. **Popularity:** Which restaurants are ordered from most often?
3. **Profitability:** Which restaurants make the most money and why?
4. **Location Trends:** How performance varies by city.
5. **Ratings & Cost:** How do rating and average cost impact success?
6. **Cuisine Influence:** What types of food dominate among top earners?

Anticipated Business Recommendations

- Promote or expand high-performing restaurants to similar cities.
- Invest in restaurants with strong ratings but low visibility.
- Identify top-performing cuisines for marketing partnerships.
- Use cost vs revenue insights to optimize pricing strategies.

Tools

- **Data Platform:** TripleTen test dataset (Zomato/Swiggy)
- **Visualization Tool:** Tableau Public

Deliverables

- **Dashboard:** Tableau Public with all visuals tied to key metrics
- **Report:** Summary of conclusions and 3–5 actionable business recommendations